Lab 4: Controller PCB Design

18-449/649 Carnegie Mellon University

Release: 10/07/2026 Due: 10/29/2026, 11:59pm Version 1.0

What you are building

In this lab you design the printed circuit board that replaces the breadboards, Nucleos and CAN modules from Labs 2 and 3 with the controller board your Lab 1 requirements document promised: two STM32F401 zones (steering and drivetrain), two independent CAN controllers and transceivers per zone, a dual-CAN section for the Raspberry Pi cockpit zone, the motor H-bridge, the servo drive, the current sensing for both, the blinkers, the self-test buttons, the status LEDs, the isolation jumpers and every test point from Labs 1 through 3. The board mounts on the car chassis, runs from a battery, and goes out for fabrication at the end of the lab.

The document is organized in the order we want you to work:

- **Part 1, high-level overview.** Power source and ratings, mechanical fit to the chassis, power metering, part selection, and the fabrication rules. Read all of it before you place a single symbol.
- **Part 2, block diagram.** The reference hardware diagram your schematic should decompose into, with the signals that cross each boundary.
- **Part 3, parts list.** The parts we order in bulk for every team. Go here first when you pick a component; it is also the list of what we can hand you if a part is late.

What we provide

Item	Why you have it
Car chassis with steering servo and two geared motors with encoders (from Lab 2)	Your board must be mounted to the car chassis with standoffs. Pick at least 3 (ideally 4) mounting holes on the chassis and measure them.
Raspberry Pi 4, supply, microSD (from Lab 2)	Cockpit zone. Connects to the board over its 40-pin header: 5 V (power from the board's 5 V rail), GND, SPI0 (MOSI, MISO, SCLK, CE0, CE1) and two interrupt GPIOs, all signals at 3.3 V.
Nucleo-F401RE boards (from Labs 2 and 3)	The on-board ST-LINK programs your STM32F401s over SWD once you remove the CN2 jumpers. Put a 4-pin SWD header on each zone.
Common parts (Part 3)	Ordered in bulk for every team. Pick from this list first; it is what we have on the shelf.
PCB fabrication order	We send the fabrication files for every team to the board house in one batch after the due date (see 1.5 for the design rules). Boards that fail the design rule check are not sent.

Part 1: High-level overview

Goal: a one-page design brief, written before the schematic, that fixes the power source and its ratings, the board outline and mounting, the current-sensing approach for each actuator, and the part choices, so that the schematic and layout in the following weeks have nothing left to decide except how. You need to get this approved by the course staff before investing more work.

1.1 Power source and ratings

The car runs untethered, so the board carries its own power input and generates every rail from it. There should be four voltage levels from one battery input at a nominal 12V: a 12V rail for the drivetrain, a 6-8V rail for the steering system, a 5V rail for anything that needs 5V (this includes the Pi), and 3V3 rail for everything else.

Ratings. The peak stalled current of the whole system is estimated to be about 10A at 12V (~120W). This accounts for all the motors being stalled and the logic being pushed to its limits. The average power draw while driving is much lower, typically under 1A, but the stall current is what sizes the components for the power input (i.e connector, reverse polarity protection, overcurrent protect). A stalled motor pulls down the power rail hard. It is recommended to put 470µF - 1000µF of bulk capacitance on the motor driver's power rail.

The easy source: AA cells in series. Eight alkaline or ten NiMH AA cells in a plastic holder give about 12 V nominal and are the simplest, safest option: no charger on the board, no protection circuit, and a flat cell is a trip to the supply closet rather than a safety incident. The caveat is internal resistance. Alkaline AAs are 150-300 mOhm each, so an eight-cell pack can sag by several volts at 5 A; NiMH cells are a few tens of milliohms each and hold up much better under stall. Either way the stall is a transient, so design the bulk capacitance and the buck input range for the sagged voltage, not the nominal one, and say in the brief what the minimum battery voltage is at which the 5 V rail still regulates. Pick a holder with a lead length that reaches your input connector where you mounted it on the chassis.

For the more adventurous: lithium. A 3S lithium-ion or LiPo pack (11.1 V nominal, 12.6 V full, about 9 V empty) has far lower internal resistance and far more energy per gram. It is also the only part of this project that can start a fire. If you choose lithium, the following are mandatory, not suggestions:

- Use a pack with a built-in protection circuit (BMS) or a hobby pack with a balance lead, and charge it only on a dedicated balance charger, off the board, in a fire-safe bag. Do not design a charger onto the PCB.
- Never discharge below 3.0 V per cell (9 V for 3S). Put a low-voltage cutoff in your drivetrain software (the ADC can watch the input through a divider) and show it working at checkoff.
- Fuse the input, protect against reverse polarity, and use a connector that cannot be plugged in backwards (XT60 is keyed; that is why it is on the list).
- Never leave a lithium pack connected to a board you are probing or reworking, and never puncture, short or charge a swollen cell. If anything smells hot, unplug it and tell the course staff.
- Follow CMU's Lithium Ion Battery Best Practices Guidelines.

Guiding Questions:

- What voltage rails should supply the L298? How about the hall-effect sensors, microcontrollers, CAN bus, LEDs, and the RPi?
- What is the stall current of one of your motors at 12 V, measured rather than looked up? What does the L298 drop at that current, and how hot does it get in 10 s of stall?
- What is the worst-case load on the 6-8V rail? What about the 5V and 3V3 rails?
- Where does the motor current return to the battery, and does that path share any copper with the ground reference of your ADC inputs or your CAN transceivers? Draw it.

Checkpoint. Your design brief should include a power tree that shows what is being powered off of each rail. Also include a power budget table (rail, source, peak current, average current, worst-case voltage) for all four rails, and a one-paragraph justification of the battery choice, including the minimum battery voltage at which every rail still regulates.

1.2 Mechanical design

The board is part of the car, not a thing that sits next to it. It should be able to be mounted to the car chassis with standard PCB standoffs through holes that line up with the chassis holes.

- **Measure the chassis.** With calipers, record the mounting hole positions relative to one reference corner, the hole diameters, the usable board area, and the height available above and below the board. Put the measurements, with a sketch, in the design brief. Pick a standoff size (M3 with 3.2 mm holes, or M2.5 with 2.7 mm holes, whichever the chassis is tapped for) and put it in your dimension drawing.
- **Keep-outs.** Before laying out your PCB, place your mounting holes on the board and start thinking where you are going to put the RPi and the battery pack. Do not place anything within 3mm of a mounting hole.
- **Connector orientation.** Motor and encoder connectors should go near the motors; the servo connector should go near the servo; the battery connector where the pack lead reaches it. Cables you cannot route are cables you will cut and re-solder at 2 a.m.
- **Reachability.** R4.8 requires the isolation jumpers to be reachable without disassembling the vehicle, and the TA will be pulling them while the car runs. The same goes for the termination jumpers (R4.3), the CAN_H/CAN_L test headers (R4.6), the self-test buttons, the test-point rows and the SWD headers. Put all of them in accessible locations and label them in silkscreen.

Guiding Questions:

- Where is the center of mass of the car once the battery and the board are on it, and does that matter for the drive test?
- Which direction do the motor leads exit, and what happens to your connector placement if the chassis is assembled mirror-image from the one you measured?
- If the Pi stacks on standoffs above the board, which components are under it, and is the tallest of them lower than the standoff?

Checkpoint. A board outline with mounting holes, keep-outs and connector positions is drawn in your layout tool.

1.3 Power metering

The board carries two kinds of current sensing, and they are deliberately different. The rear drive motors keep the hall-effect sensors defined in Lab 2; the steering servo moves to a shunt-resistor design. The point of doing both is that you will have measured both on the same board and can argue about which one you would use in different situations.

Rear motors: ACS712 hall-effect sensors, as in Lab 2 (R6.1). One ACS712ELCTR-05B per motor, in series with one motor lead between the H-bridge output and the motor connector, so the sensor sees the bidirectional motor current directly. Bring the sensor output to a test point (R6.5).

Steering servo: shunt resistor. The force-feedback value (Lab 1 Front Zone, R7.3) comes from servo load, and servo load at small steering corrections is a few tens of milliamps on top of the idle current. Use a shunt resistor to measure the current of the servo motor. When selecting a shunt resistor, keep in mind the servo's stall current and the resistor's power rating. Also consider where to place the sensing circuit (i.e low side, high side, in line), the ADC configuration (i.e single-ended vs differential), and if you need an amplifier. Place an RC filter near the ADC pin(s) with a corner well above the servo's PWM frame rate and below the ADC's sample rate. Bring the unfiltered ADC input(s) to a test point(s). STMicroelectronics, Texas Instruments, and others all have decent guides to current sensing.

Guiding Questions:

- What is the idle current of your servo holding position with no load, and what is the smallest change in load you want to feel on the steering wheel? What shunt value and gain does that imply?
- The servo draws a pulsed current at its internal PWM frequency. What does your ADC see if you sample without a filter, and what averaging do you do in software?
- The ACS712 has a 2.5 V zero-current offset that drifts with temperature. How do you calibrate it at power-up when the motor is guaranteed to be stopped?
- What are the advantages and disadvantages of high side current sensing? Low side current sensing? In line current sensing?

Checkpoint. Your design brief has, for each of the three sensors, the full-scale current, the scale factor in mA per ADC count, the resistor power rating, the filter corner and the test point names.

1.4 Part selection

Pick parts in this order, and write down why when you deviate:

1. The common parts list in Part 3. We order these in bulk for every team, so they are on the shelf, we know their footprints work, and a lost or damaged part is replaced the same day. If a list part does the job, use it. If the list says "Low stock" or "No" under JLCPCB stocking, we will still have it on hand; it just will not be machine-placed (see below).

2. JLCPCB assembling option. Depending on timing, we might be able to get JLCPCB to assemble some of the main components for you right when they make the PCB. This can be good if you are not skilled at soldering, but comes at the cost of making it hard to incrementally bring up your board. If you would like JLCPCB to assemble your PCB for you, you should base your BoM off of JLCPCB's in-stock components library. Use parts that they have on hand already and prefer **Basic** parts over **Extended** parts. Basic parts are always loaded on the assembly line and cost nothing extra, and every extended part adds a per-part loading fee and may be out of stock when we order (increases lead time of your PCB).

3. Anything else. A part that is not on the list is one you order from Digikey or Mouser. Be sure to select parts that have plenty of stock available and can be purchased in smaller quantities.

Some part-level guidance that comes up every year:

- **MCU.** STM32F401RET6 is the Nucleo's chip, so your Zephyr code, overlays and the Nucleo's ST-LINK (over SWD) all carry over. Refer to the Nucleo C03 CAD resources as a starting point.
- **CAN.** Six MCP2515 + SN65HVD230 pairs. Each MCP2515 needs its own 16 MHz crystal and load capacitors, or its sibling's CLKOUT within the same zone. The transceivers on the end of the bus gets a 120 Ohm termination, placed on the bus side of the isolation jumper (R4.3), a 2x2 header or DIP switch that breaks both CAN_H and CAN_L (R4.8), and a CAN_H/CAN_L test header (R4.6). The RS pin of the SN65HVD230 sets slope control; strap it per the datasheet for 500 kbit/s and say what you chose.
- **H-bridge.** The L298 is the IC that drives your motors, and since it drives your motors, you should think about effectively dissipating its heat. The powerSO package requires an exposed pad and sinks its heat through the PCB. However, if you end up using the Multiwatt-15 package, you will need a heatsink to effectively dissipate the heat generated.
- **Passives.** Use 0603 packages or larger for all the passive components. It is discouraged to use 0402 components, and any components smaller than 0402 really are not needed.
- **Connectors and headers.** JST-PH 2.0 mm (list) for motor, encoder and servo pigtails, matching what is on your chassis cables; 2.54 mm headers for everything the TA touches. Label every header in silkscreen with its signal names, not just a reference designator.
- **Test points.** Test points must be placed on the drivetrain PWM signals, the steering PWM signal, all the blinkers, the self test buttons, the error state LEDs, and the UDP reception GPIO. Additionally, you need to place a few ground test points and test points on each power rail. These test points should be labeled on the silkscreen. Other test points can be added at your discretion and do not need to be easily accessible (since they aren't checked by the TAs). The

test points the TAs need to observe must be 2.54mm headers or Keystone 5015 (list) test point hooks. Any other test points can be exposed copper pads.

1.5 Fabrication: JLCPCB and layer count

We send every team's board to JLCPCB in one order, so your layout must pass JLCPCB's design rule check as submitted. Set your design rules from JLCPCB's published capabilities (jlcpcb.com/capabilities/pcb-capabilities) before you route, not after. The numbers that matter for a 2-layer, 1 oz board are below; check the page for the current values, and use the JLCPCB rule file for your tool (KiCad and EasyEDA both ship one) rather than typing them in by hand.

Rule (2-layer, 1 oz)	JLCPCB minimum	Use at least	Why
Trace width	0.127 mm (5 mil)	0.2 mm signals, 0.5 mm+ power, pours for motor current	Yield; 5 A needs ~3 mm
Trace-to-trace / trace-to-pad clearance	0.127 mm (5 mil)	0.2 mm	Solder bridges on hand-soldered boards
Via drill / via diameter	0.3 mm / 0.5 mm	0.3 mm / 0.6 mm	Annular ring margin
Via-to-via, via-to-track	0.254 mm	0.3 mm	DRC
Copper to board edge	0.3 mm	0.5 mm	Routing tolerance
Plated hole for standoffs	0.3 mm min hole	3.2 mm (M3) with 6 mm pad	Standoff fit
Silkscreen line / text height	0.153 mm / 1 mm	0.2 mm / 1.2 mm	Legibility
Layers	2, 4, 6	2 expected; 4 maximum	Cost and lead time

This board should be routable on two layers: a top layer for signals and the power paths, and a bottom layer that is mostly a ground plane with a few escape routes. Keep the ground plane under every CAN pair, every SPI bus and every ADC input, and route each CAN_H/CAN_L pair as a coupled differential pair, physically separated from the other bus (R4.3). Keep the motor return current out of the logic ground plane with a single star point. If you cannot close the layout on two layers, four layers (signal, ground, power, signal) is permitted and makes the ground plane trivial, but it costs more, takes longer, and we will ask you at checkoff why two did not work. More than four layers is not permitted.

Guiding Questions:

- What is the controlled-impedance target for a CAN pair, and does a 2-layer 1.6 mm board let you hit it? Does it matter at 500 kbit/s over 30 cm?
- What is the minimum annular ring JLCPCB will build, and what does your via of 0.3 mm drill and 0.6 mm diameter leave as margin for drill wander?

Checkpoint. Your layout tool's design rules are set from the JLCPCB rule file, the stackup is 2 layers (or a justified 4), and a DRC run on the empty board with outline and mounting holes reports zero errors.

Part 2: Block diagram

Figure 2 is the reference hardware diagram. Your schematic should decompose into exactly these blocks, one schematic sheet per zone plus a power sheet, and a reader who knows this figure should be able to find any signal on your schematic in under a minute. You may add blocks (a collision sensor, a buzzer, a second input divider) and you may substitute within a block (a different H-bridge, a different buck), but every block that is here must be on your board, and the three zones must stay electrically independent except for the two CAN buses, ground, and power rails.

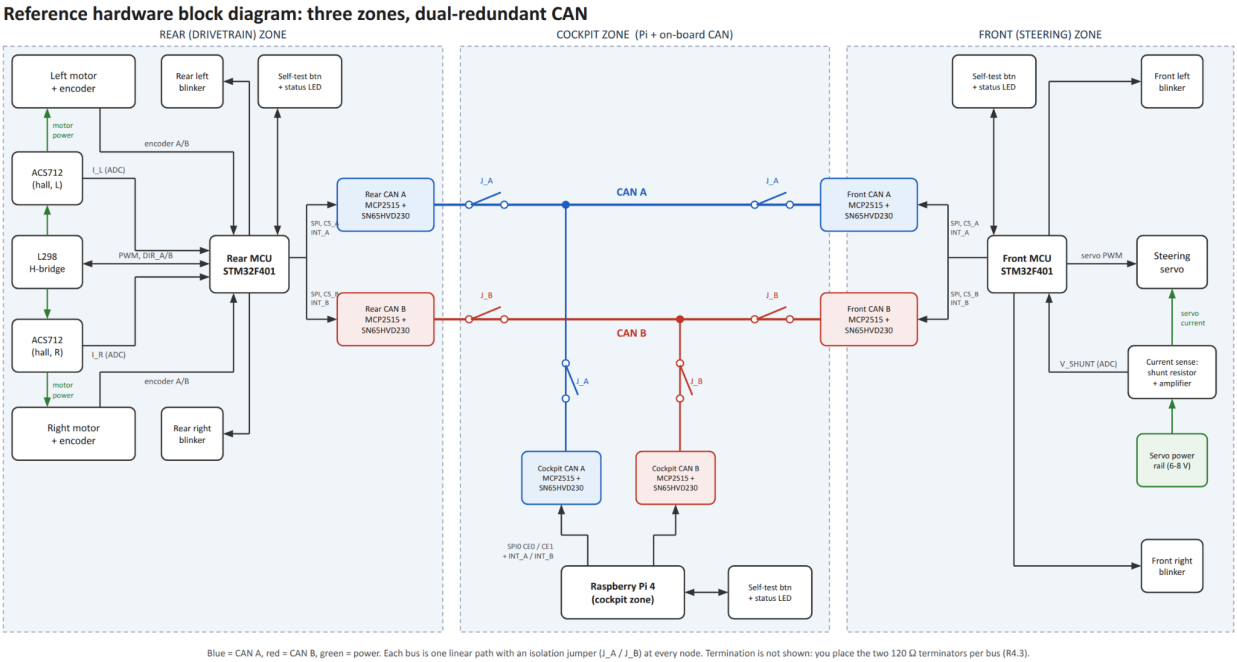


Figure 1: Reference hardware block diagram

Reading the figure: blue is CAN A, red is CAN B, green is power. The two STM32 zones are on the left and right, the cockpit zone in the middle with the Pi at the bottom; the two buses run horizontally between the zones. Each node reaches each bus through its own MCP2515 and SN65HVD230, then through an isolation jumper (J_A, J_B) that breaks both CAN_H and CAN_L, onto a single linear bus with one 120 Ohm terminator at each end. The terminators sit on the bus side of the jumpers, so pulling any node, including an end node, leaves both terminators in place (R4.3). The cockpit zone's two CAN pairs are on your board and connect to the Pi over SPI0 with two chip selects and two interrupt lines, exactly as in Lab 3 Appendix B. The rear zone drives both motors through the L298 and reads each motor's current through its ACS712; the front zone drives the servo from its 6 V rail and reads the servo current through the shunt resistor. Both STM32 zones, and the RPi have a self-test button, status LEDs, and the per-bus degraded LEDs of R4.7.

Requirements checklist. Before the design review, tick every item against your schematic and layout; the TA will.

Requirement	What the board must have
R1	Two STM32 zones with independent clock sources (own crystal or own internal oscillator); no clock net crosses a zone boundary.
R2.5 / R3	A hardware self-test button per STM32 zone, debounced in hardware (RC) or software, and an ERR output that is also a test point.
R4.1	Two MCP2515 + SN65HVD230 pairs per zone (six total), with no shared transceiver, controller or bus wiring between A and B.
R4.2	16 MHz crystal (or sibling CLKOUT) for every MCP2515 so all nodes can run identical 500 kbit/s bit timing.
R4.3	Each bus a single linear path across the board; 120 Ohm termination footprints with selection jumpers at both ends, bus-side of the isolation jumpers; CAN_H/CAN_L routed as a differential pair, A and B physically separated; distinct, labeled connectors for A and B.
R4.6	CAN_H/CAN_L test header or DIP switches for both buses at every zone; CAN_TX and CAN_RX test points per bus per zone.
R4.7	DEG_A and DEG_B LEDs on every node (the cockpit's two live on the board and are driven by Pi GPIOs).
R4.8	Isolation jumpers that break both CAN_H and CAN_L, per bus, per node, between transceiver and bus connector, labeled, reachable without disassembly.
R5.1 / R6.1 / R6.5	Encoder inputs for both motors; ACS712 per motor into an ADC pin; current-sense outputs on test points; overload GPIO test point.
R7.3	Servo load sensing (shunt) into an ADC pin with a test point.
Lab 2 / Lab 3 test points	Every row of the Lab 3 test-point table, per zone, in the same order, with two ground pins per group.

Guiding Questions:

- Which signals are 5 V and which are 3.3 V?
- Where does each zone's ground reference enter the ground plane, and does the motor return current flow between them? What does that do to the ADC readings on the ACS712s?
- Which Pi GPIOs did you reserve?

Checkpoint. A schematic for Figure 1 (rear zone, front zone, cockpit CAN, power), every net in the interface table named, every checklist row ticked, and an ERC with zero errors.

Part 3: Parts list

These are the common parts we order in bulk for every team. Go here first. The JLCPCB column says whether the part is in JLCPCB's assembly library at the time of writing; "Yes" means it can be machine-placed if we have time for assembly, "Low stock" or "No" means we will have it on the shelf for you to hand-solder. Part numbers link to DigiKey (and Mouser where it matters). Quantities are per board; order a few spares of anything you will hand-solder.

Description	Part number	Manufacturer	Package	Use on this board / notes
Buck switching regulator IC, adjustable, 4 A output	TPS564252DRLR	Texas Instruments	SOT-563 / SOT-666	Stuff
CAN controller, CAN 2.0, SPI interface	MCP2515T-I/ST	Microchip Technology	20-TSSOP	Six per board (two per zone). Each needs its own 16 MHz crystal or a CLKOUT feed from a sibling in the same zone.
CAN transceiver, 3.3 V	SN65HVD230MDREP (Mouser)	Texas Instruments	8-SOIC	Alternative: SN65HVD230DR (same pinout). Six per board.
Dual full-bridge driver (DC motors)	L298P (Mouser)	STMicroelectronics	PowerSO-20, exposed pad	Alternative: L298N (Multiwatt-15 through-hole, same die). Low stock everywhere; may need a heatsink.
STM32F4 Cortex-M4 MCU, 84 MHz, 512 KB flash	STM32F401RET6TR	STMicroelectronics	64-LQFP (10x10)	Same part as the Nucleo-F401RE, so your Lab 2/3 Zephyr code ports directly.
32.768 kHz crystal, 6 pF	ABS25-32.768KHZ-6-T	Abracon LLC	4-SOJ, 5.5 mm pitch	Optional. Only needed if you want the RTC / LSE
16 MHz crystal, 12 pF, 80 Ohm	CX3225SB16000H0FLJCC	KYOCERA AVX	4-SMD 3.2x2.5 mm	3225 16 MHz crystal
Schottky diode, 200 V, 5 A	SK520B	SMC Diode Solutions	DO-214AA (SMB)	Stuff
Tactile switch, SPST-NO, top actuated, SMD	SWT0210-045026GSA	GCT	SMD	Self-test buttons (one per STM32 zone) and reset. Any NO tactile pushbutton works.
Hall-effect current sensor, +/-5 A, bidirectional	ACS712ELCTR-05B-T	Allegro MicroSystems	8-SOIC	Rear motor current sensing, one per motor, as in Lab 2.
120 Ohm 1 % 0603 resistor	RC0603FR-07120RL	YAGEO	0603	CAN termination, two per bus. Any 120 Ohm 1 % part works.
2.54 mm pin headers (breakable strips)	-	-	Through-hole	Standard through-hole pin headers

Description	Part number	Manufacturer	Package	Use on this board / notes
Connector header, SMD, 6 position, 2.0 mm (JST PH)	B6B-PH-SM4-TB	JST Sales America Inc.	SMD	DC Motor connections to your PCB.
Yellow LED, 590 nm, 3 mm radial	151031YS06000	Wurth Elektronik	3 mm round, through-hole	Blinkers. Chip LEDs are fine if you prefer.
Red LED, 640 nm, 3 mm radial	151031SS06000	Wurth Elektronik	3 mm round, through-hole	Error / degraded-bus status. Chip LEDs are fine if you prefer.
22 Ohm 5 % 0603 resistor	CRGCQ0603J22R	TE Connectivity	0603	Stuff
XT60 male connector, gold plated	FIT0587	DFRobot	Through-hole	Battery input. A screw terminal or 5.5/2.1 mm barrel jack is an acceptable alternative for AA packs.
PC test point, miniature, SMD	5015	Keystone Electronics	SMD	Every test point required in Labs 1-3. Nice for debugging with scope clips.
0.1 µF ±10% 25V Ceramic Capacitor X7R 0603 (1608 Metric)	CGA3E2X7R1E104K080AA	TDK	0603	Decoupling
4.3 pF ±0.25pF 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	CC0603CRNPO9BN4R3	YAGEO	0603	Crystal load capacitors
22 pF ±5% 100V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	C0603C220J1GACTU	KEMET	0603	Crystal load capacitors
10 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Automotive AEC-Q200 Thick Film	CRCW060310K0FKEA	Vishay Dale	0603	Stuff
600 Ohms @ 100 MHz 1 Signal Line Ferrite Bead 0603 (1608 Metric) 500mA 350mOhm	742863160	Würth Elektronik	0603	AVDD filtering
470 µF 25 V Aluminum Electrolytic Capacitors Radial, Can 2000 Hrs @ 105°C	860020474014	Würth Elektronik	Through Hole	Bulk capacitance

Description	Part number	Manufacturer	Package	Use on this board / notes
47 kOhms ±1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resistant Thick Film	RC0603FR-0747KL	YAGEO	0603	Stuff
4.7 µF ±10% 25V Ceramic Capacitor X5R 0805 (2012 Metric)	GRM219R61E475KA73D	Murata Electronics	0805	Decoupling

Not on the list, pick from JLCPCB. Any components that are not on the list can be purchased from digikey or mouser. If you choose to do JLCPCB in-stock stuffing, choose from their parts list.

Ordering timeline. By October 21, please send your TA your BoM. This needs to include the part number, manufacturer, quantity (plus a little extra), and the distributor (digikey or mouser). This is so we can place the order before the board arrives. Also tell your TA if you intend to have JLCPCB assemble your PCB otherwise.

Design review

Every team meets with their TA for a 1-hour design review during the week of October 19. Bring the design brief from Part 1, the schematic from Part 2 and the current state of the layout. The review is graded (below) and is the last chance to catch a mirrored connector or a missing jumper before fabrication. Teams whose review finds a blocking issue get one re-review slot before the due date.

If you would like a more detailed review, please send Tom your schematics and layout. He is very experienced in PCB design and will tear your design apart.

Deliverables

Submit to Gradescope by 10/29/2026, 11:59pm. We send the fabrication files to JLCPCB in one batch the next morning; a submission that fails the JLCPCB DRC or is missing a file is not sent, and the lab is not complete until it is.

Item	Points
Design review with TA (week of 10/19): brief, schematic and layout in hand, blocking issues resolved	10
Design brief (Part 1): power budget and battery justification, chassis measurements and outline drawing, current-sensing design for all three sensors, part choices with reasons	15
Schematic (PDF, one sheet per block): every interface-table net named, every checklist requirement present, ERC clean, every part annotated with value, footprint and LCSC number	25

Item	Points
Layout (PDF of each layer plus a 3D render or photo of the 1:1 print on the chassis): outline and holes match the chassis, keep-outs respected, CAN pairs differential and separated, ground plane and star point, headers reachable and labeled	25
Fabrication files: Gerber + drill zip that passes the JLCPCB DRC with zero errors, the DRC report, layer count stated	15
BOM (with LCSC C-numbers and Basic/Extended flag) and CPL in JLCPCB format, plus the list-part quantities you need from the shelf	10
Lab 1 revision-history entry recording what the board changes relative to your Lab 1 hardware block diagram and message catalog (pins, IDs, LED assignments)	Required

The brief should be no more than four pages excluding figures and the chassis drawing. Tables of numbers are expected; prose should be the reasoning that connects them. Every current, voltage and dimension you use must say whether it was measured or taken from a datasheet, and a measured number must say how.

Appendix A: Things that often go wrong

Symptom	Usually means
Board does not fit, standoffs do not line up	Chassis measured from a datasheet drawing, or holes placed from a corner that is not square. Measure the real chassis; print 1:1 before ordering.
6 V or 5 V rail drops out when a motor or the servo stalls	Battery sag plus no bulk capacitance; buck input below its minimum, or the servo sharing a rail with logic. Check the sagged battery voltage, not the nominal, and keep the servo on its own buck.
Servo is sluggish or chatters, or runs hot	Fed from 5 V (below its 6 V minimum) or from 12 V (above its 8.4 V maximum). The LD-1501MG wants its own 6-8V zone.
Pi reboots when the motors start, or will not boot from the board	No bulk capacitance.
Servo twitches, force value is noisy	No RC filter on ADC input; No filtering in firmware or software.
JLCPCB DRC fails on clearance or annular ring	Design rules set after routing, or a library footprint with tiny pads. Load the JLCPCB rule file first and run DRC on the empty outline.
Assembly quote is expensive, or parts are out of stock when we order	Many Extended parts, or SMD on both sides. Prefer Basic parts and one-sided SMD; record C-numbers from day one.
CAN works on the bench, fails on the car	Bus A and B routed as single-ended traces next to a motor trace, no ground plane under them, or termination on the node side of the jumper. R4.3 and the layout guidance in 1.5.

Symptom	Usually means
STM32 will not program	SWD header wired wrong, BOOT0 floating, NRST not brought out, or the Nucleo's CN2 jumpers still in place. Follow the Nucleo user manual for external SWD.
One zone's CAN bit rate is wrong	MCP2515 crystal is a different frequency from the devicetree osc-freq. All six should match; say what they are in the brief.
Isolation jumper pulled and the whole bus dies	Terminator on the node side of the jumper, or the jumper breaks only one of CAN_H / CAN_L. Both must break; terminator stays.

Appendix B: Common Protection Circuits

This section provides some tips about reverse polarity protection, overcurrent protection, switching inductive loads, etc.

Reverse Polarity Protection:

There are a few different ways to protect your design against reverse polarity. One viable option is to select a regulator that protects against it for you. However, that is not always an option, especially if you are running things off the input voltage directly, or if you have polarized capacitors on your input. There are a few circuits you can employ:

Diode: The simplest form of reverse polarity protection is a diode, and more specifically a schottky diode. You put the diode in line on the high side and you're done. It's a very simple and cheap form of reverse polarity protection, but once you start drawing more than 2 or 3A, you start running into power loss issues (schottky diodes are more power efficient than regular diodes).

P-channel MOSFET: Another way to protect against reverse polarity is by placing a P-channel MOSFET on the high side. This is a popular option for applications exceeding 3A. When the circuit is connected correctly, the P-channel MOSFET is open and allows current to flow. The channel voltage experiences a small drop, then the temperature rises and the overall power loss is reduced. When the power supply is reversed, the P-channel MOSFET is closed, and the parasitic body diode accomplishes the reverse polarity protection function. The down side with this circuit is the fact that it is higher cost.

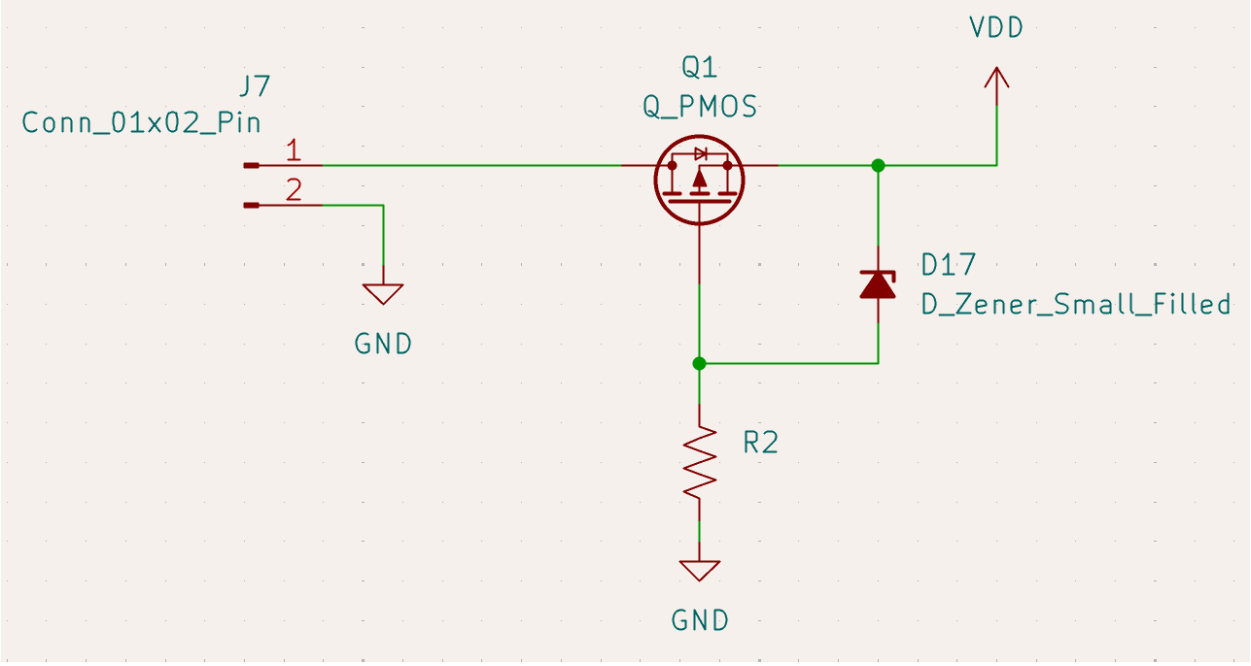


Figure 2: Reverse Polarity Protection with a P-channel MOSFET

Also note that there are two other components associated with this circuit. The zener diode protects the P-channel MOSFET from V_{ds} overvoltage, and the resistor limits how much current can flow through the zener diode. The resistor is usually in the range of 1k to 100k Ohms.

N-channel MOSFET: Like P-channel MOSFETs, N-channel MOSFETs can be used to protect against reverse polarity as well. The difference is where you place the circuit: on the low-side. This method is less popular since you are splitting your power ground and your load ground.

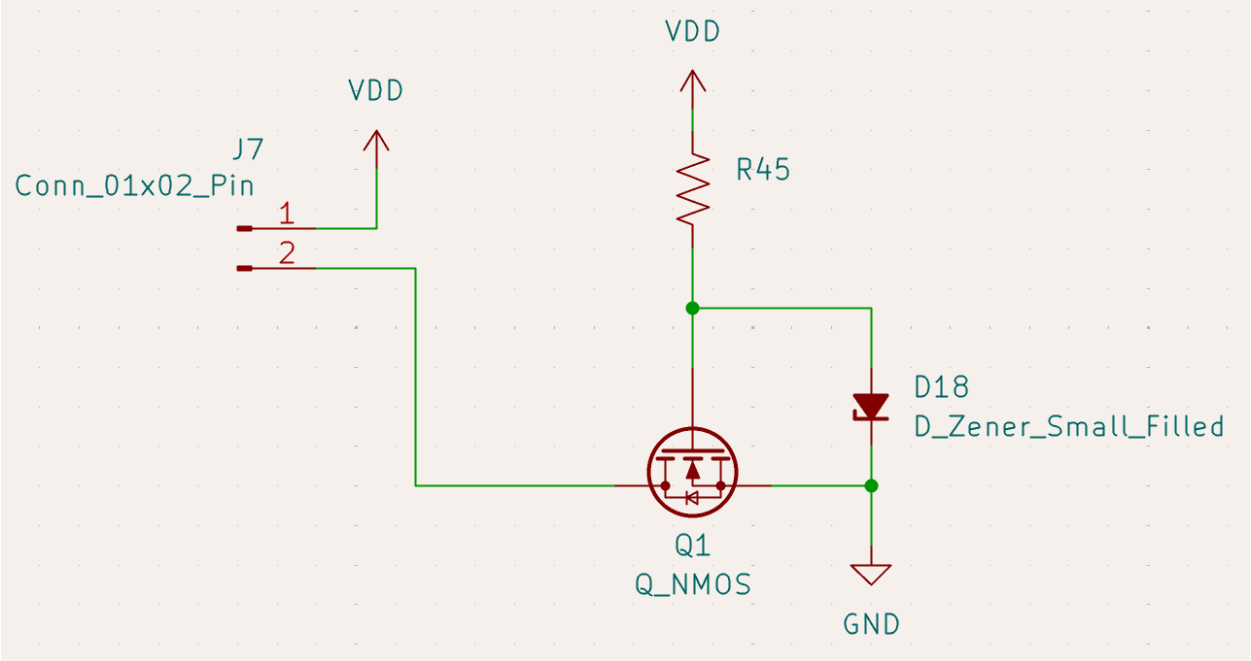


Figure 3: Reverse Polarity Protection with an N-channel MOSFET

Overcurrent Protection

The best way to protect against overcurrent is to use a fuse. You can either use a traditional fuse or a PTC fuse to protect your PCB. You generally want to pick something that is a little above your maximum current draw (1.5x).

Switching Inductive Loads

When running current through an inductive load, energy will be stored in the electromagnetic field. If you suddenly cut the current, the flyback voltage from the inductive load can be thousands of volts and can damage the switching transistor. To protect against these events, a flyback diode is necessary. The flyback diode will keep current flowing through the inductive load and helps dissipate the stored energy.

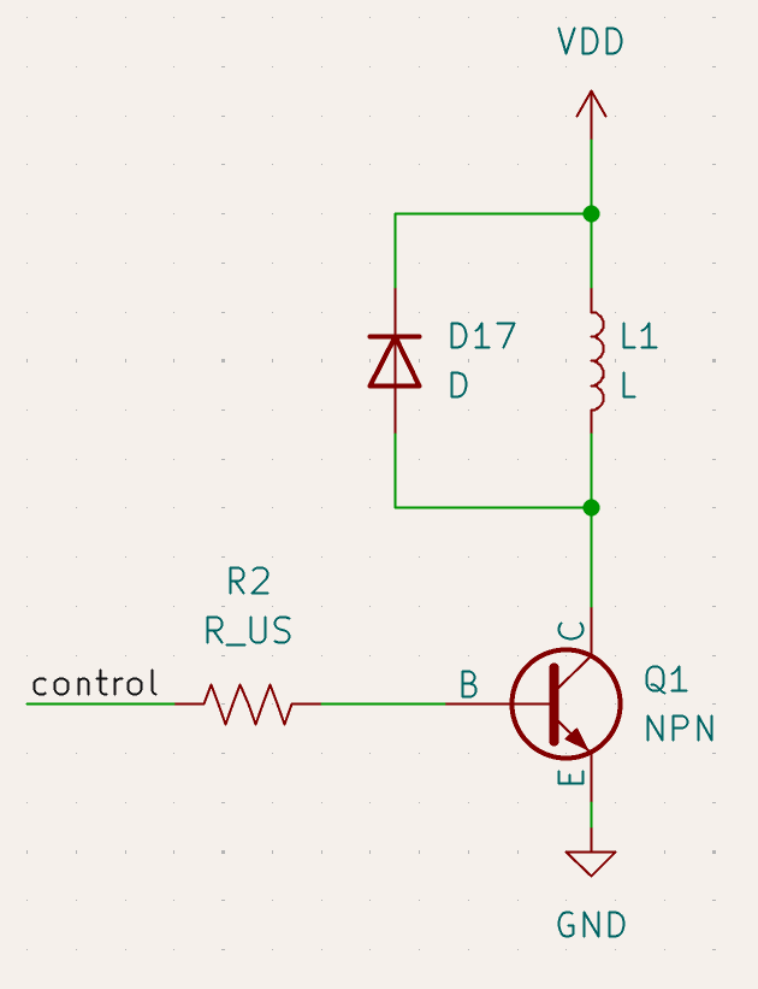


Figure 4. Switching Inductive Load

Appendix C: Popular CAD Tools

We do not require a specific design tool. In class (and 349) we used Fusion 360 which is free for students. Altium is a Windows-only, but professional option with some educational / hobby access. KiCAD is another good open-source alternative.

Revision History

10-09-26 Version 1.0: Initial release to the class